The familiar extended dimensions, therefore, may very well also be in the shape of circles and hence subject to the R and 1/R physical identification of string theory. To put some rough numbers in, if the familiar dimensions are circular then their radii must be about as large as the 15 billion light-years mentioned above, which is about ten trillion trillion trillion trillion trillion (R = 1061) times the Planck length, and growing as the universe expands. If string theory is right, this is physically identical to the familiar dimensions being circular with incredibly tiny radii of about 1/R=1/1061 = 10-61 times the Planck length! These are our well-known familiar dimensions in an alternate description provided by string theory. In fact, in this reciprocal language, these tiny circles are getting ever smaller as time goes by, since as R grows, 1/R shrinks. Now we seem to have really gone off the deep end. How can this possibly be true? How can a six-foot tall human being "fit" inside such an unbelievably microscopic universe? How can such a speck of a universe be physically identical to the great expanse we view in the heavens? Furthermore, we are now led forcefully to the second of our initial three questions: String theory was supposed to eliminate the ability to probe sub-Planck distances. But if a circular dimension has radius R whose length is larger than the Planck length, its reciprocal 1/R is necessarily a fraction of the Planck length. So what is going on? The answer, which will also address the first of our three questions, highlights an important and subtle aspect of space and distance.

Two Interrelated Notions of Distance in String Theory

Distance is such a basic concept in our understanding of the world that it is easy to underestimate the depth of its subtlety. With the surprising effects that special and general relativity have had on our notions of space and time, and the new features arising from string theory, we are led to be a bit more careful even in our definition of distance. The most meaningful definitions in physics are those that are operational—that is, definitions that provide a means, at least in principle, for measuring whatever is being defined. After all, no matter how abstract a concept is, having an operational definition allows us to boil down its meaning to an experimental procedure for measuring its value.